

Team 1

Rubber Duck Programmers

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Design 1: Phone Application

Design Overview

Our first design is a mobile application designed to assist individuals with impaired memory in managing their medications while allowing caregivers to provide support when necessary. The application contains two separate interfaces: one for the dependent taking the medication and another for their caregiver.

The dependent interface prioritizes simplicity, accessibility, and independence. Rather than displaying large amounts of information at once, the application focuses on what the user needs to do at that moment. The caregiver interface provides more detailed information and allows caregivers to manage medication schedules, monitor medication status, and provide assistance when needed.

Dependent Interface

The dependent's home screen displays their next scheduled medication. The screen includes the medication name, a picture of the medication or bottle, dosage, scheduled time, and basic instructions. Large text, simple language, consistent button placement, and medication images are used to make the interface easier to understand.

When it is time for medication, the dependent receives a visual and audio reminder. Caregivers can also record personalized audio reminders so that the dependent can hear instructions or encouragement from a familiar person.

The dependent is given three primary options when a medication reminder appears:

- **Take Medication**
- **Remind Me Later**
- **I Need Help**

Selecting "Take Medication" begins the medication verification process. Selecting "Remind Me Later" temporarily dismisses the reminder without marking the medication as taken. "I Need Help" allows the dependent to easily contact an approved caregiver if they are confused or need assistance. The dependent can also view their medication schedule for the day, including medications that have already been taken, upcoming medications, and medications that still require attention.

Medication Verification

The application uses a two-step verification process to reduce medication errors. Before taking medication, the dependent scans the medication bottle using their phone's camera. The application compares the scanned medication with the medication scheduled for that time. If the correct medication is scanned, the application displays the dosage and instructions and allows the dependent to continue.

If the incorrect medication is scanned, the application clearly tells the dependent not to take it and displays the medication they should be looking for. The warning does not rely on color alone and instead uses text, symbols, medication images, and clear instructions.

After taking the medication, the dependent completes a second verification by confirming that the medication was taken. Only then is the medication recorded as **Taken**. Scanning a medication alone does not count as taking it. If either verification step is incomplete, the medication remains **Unconfirmed**. The application can continue reminding the dependent and eventually alert the caregiver if the medication is not confirmed.

Caregiver Interface

The caregiver interface provides more detailed medication management and monitoring tools. Caregivers can add medications, dosages, instructions, scheduled times, and photographs of medication bottles or pills. They can also edit existing medication information and record personalized audio reminders for the dependent.

The caregiver dashboard displays the dependent's medications using three primary statuses:

- **Taken**
- **Upcoming**
- **Missed/Unconfirmed**

Caregivers can view medication history and receive notifications when a dependent does not complete the medication verification process. They can also contact the dependent through the application if assistance is necessary. The account can support multiple approved caregivers or family members, allowing access to be shared between people involved in the dependent's care.

Optional Location and Safety Features

The application also includes an optional location-sharing system intended to provide additional safety support. If enabled by the dependent, approved caregivers can view the dependent's current or last known location, the time the location was last updated, and device battery information when available. Optional notifications can alert caregivers when the dependent arrives at or leaves predefined locations.

Location sharing is not required to use the medication-management features. The dependent can control whether location sharing is enabled and which approved caregivers can access their location. The interface also clearly indicates when location sharing is active. This feature is intended to provide additional assistance for individuals who may become lost or disoriented while still allowing the dependent to maintain control over their privacy.

Example Interaction

A caregiver initially enters the dependent's medication information, including the medication name, dosage, schedule, instructions, and a photograph of the bottle. The caregiver may also record a personalized reminder. When the medication is due, the dependent receives an audio and visual

notification. The home screen displays the medication and provides the options to take it, receive another reminder later, or request help.

If the dependent selects “Take Medication,” they are asked to scan the medication bottle. The application verifies that the bottle matches the scheduled medication. If it is incorrect, the dependent is warned not to take the medication and is shown the correct medication

After successfully verifying the bottle, the dependent receives the dosage instructions. Once the medication has been taken, the dependent confirms completion. The medication is then marked as Taken, along with the confirmation time, and the caregiver’s dashboard automatically reflects the updated status. If the dependent does not complete the process, the medication remains Unconfirmed. The application continues providing reminders and can eventually notify the caregiver that assistance may be necessary.

Strengths

- Supports the dependent’s independence rather than requiring constant caregiver involvement.
- Provides two-step medication verification to help reduce medication mistakes.
- Uses medication photographs and simple instructions to assist with identification.
- Provides familiar audio reminders recorded by caregivers or family members.
- Allows caregivers to monitor medication status remotely.
- Provides an easily accessible method for the dependent to request help.
- Offers optional location sharing as an additional safety feature.
- Can support multiple caregivers or family members.
- Uses accessibility features such as large text, large buttons, simple language, audio instructions, high contrast, and information that does not rely solely on color.

Weaknesses and Limitations

- Requires the dependent to own and regularly carry a compatible smartphone.
- The dependent may forget where their phone is or forget to charge it.
- Users with significant cognitive or physical impairments may still have difficulty navigating a smartphone.
- Scanning the correct medication does not guarantee that the medication was actually taken.
- Final confirmation still depends on the dependent accurately reporting that the medication was taken.
- Camera-based medication verification may fail because of damaged labels, lighting, or camera problems.
- Location sharing introduces privacy concerns and requires clear user consent and access controls.
- The application depends on caregivers correctly entering medication and scheduled information.



Design Self-Check

What makes this design fundamentally different from the other two designs?

The phone application differs primarily through its interaction modality and context of use. Unlike the wearable sleeve, which provides quick and glanceable information through a device worn on the body, the application provides a more detailed touchscreen interface with multiple functions. Unlike the smart medication station, the application is portable and does not physically store or dispense medication.

The application also emphasizes remote communication between two stakeholder groups. The dependent uses the application for medication guidance and verification, while caregivers use a separate interface for medication management and monitoring.

What assumptions does this design make about the user and their situation?

The design assumes that the dependent has access to a smartphone and retains enough ability to perform basic interactions such as reading simple instructions, pressing large buttons, and using the phone's camera. It also assumes that a caregiver or family member is available to initially configure medication information and respond when assistance is needed.

The design assumes that the dependent may experience memory difficulties but should maintain as much independence and control as possible. Because of this, caregiver involvement primarily occurs when the dependent requests help or when the application identifies that a scheduled medication remains unconfirmed.

How does the interaction flow differ from the other designs?

The phone application uses a multi-step touchscreen interaction. The dependent receives a reminder, chooses how to respond, verifies the medication using the camera, follows medication instructions, and confirms completion.

The wearable sleeve instead emphasizes short, glanceable interactions with minimal input, while the smart medication station provides assistance from a fixed physical location and can directly organize or dispense medication. The phone application therefore requires more active interaction but provides greater flexibility and access to additional features.

How is this design informed by Milestone 1?

This design responds to the needs and design requirements identified during Milestone 1 by emphasizing medication identification, reminders, accessibility, caregiver support, and the dependent's ability to maintain independence. The separation between the caregiver and dependent interfaces allows each stakeholder to receive information appropriate to their role rather than presenting both users with the same interface.

The dependent interface minimizes unnecessary navigation and emphasizes the next required action, while the caregiver interface provides more detailed management and monitoring capabilities. Medication photographs, audio reminders, large interface elements, and multiple forms of feedback were included to support users who may experience memory or accessibility challenges.

Design 2: Wearable Arm Sleeve

Design Overview

Our wearable arm sleeve is designed to be worn by a person with impaired memory. The design focuses on providing a simple way to manage medication and support the user without relying on several screen interfaces or requiring significant input from the wearer.

The sleeve is designed to require as little interaction from the wearer as possible while still being able to actively monitor and support them. This is intended to make the technology less intimidating for individuals who may be uncomfortable or unfamiliar with technology. It also reduces the amount the user must remember to do. Once the sleeve has been put on, the user does not need to remember to carry or regularly check another device. The sleeve primarily communicates with the wearer through a small screen, speakers, vibrations, and a physical confirmation button.

Screen

The sleeve features a simple screen that displays information about the medication the wearer is scheduled to take. This includes an image of the medicine bottle, the medication dosage, and basic instructions. When the wearer has taken their medication, they confirm this by pressing a physical button on the sleeve. This minimizes the number of screens or menus that the wearer needs to navigate. If the confirmation button is not pressed within a designated period, the system sends an alert to the caretaker's phone indicating that the medication has not been confirmed.

The screen can also be used by an authorized caretaker or doctor to input or modify medication and dosage information. These settings could be password protected to prevent the wearer from accidentally or intentionally changing important medication information.

Speakers

The sleeve is equipped with speakers that verbally announce when it is time for the wearer to take their medication. The speakers can also play recorded messages from family members or caretakers. Using familiar voices may help make reminders more recognizable and comfortable for the wearer compared with a generic alarm. Speaker volume can be adjusted using the sleeve's screen to ensure that announcements and messages can be heard by the wearer.

Directional Vibrators

The sleeve contains a ring of vibration devices that provide both medication reminders and directional assistance. When it is time for the wearer to take medication, the sleeve can vibrate in addition to providing the visual and audio reminder. This gives the wearer multiple ways of noticing that medication is due without requiring them to regularly check the screen.

The vibration system can also provide directional assistance if the wearer leaves a designated area. When the system detects that the wearer has left this area, an alert is sent to the caretaker. The caretaker can then choose whether to activate directional vibrations or ignore the alert. When activated, the sleeve vibrates on a specific side to indicate the general direction the wearer should travel to return home. This allows the wearer to receive guidance without needing to interpret a traditional map or navigate multiple screens.

Example Interaction

At the beginning of the day, the wearer puts on the sleeve. Once it is being worn, most functions occur automatically without requiring the wearer to navigate menus or remember to check the device. When it is time to take medication, the sleeve vibrates and plays an audio announcement. The screen displays an image of the medication bottle along with the correct dosage. The wearer locates the medication, takes the required dose, and presses the confirmation button on the sleeve.

After the button is pressed, the sleeve records the medication as confirmed. If the wearer does not press the button within the designated period, the sleeve sends an alert to the caretaker's phone so that the caretaker can check on the wearer.

In a second possible scenario, the wearer leaves a designated safe area. The sleeve detects that the wearer has left the area and sends an alert to the caretaker. If the caretaker determines that the wearer needs assistance returning home, they can activate the directional vibration system. The sleeve then provides vibration cues indicating the direction the wearer should travel. These interactions allow the sleeve to provide active support while requiring relatively little direct interaction from the wearer.

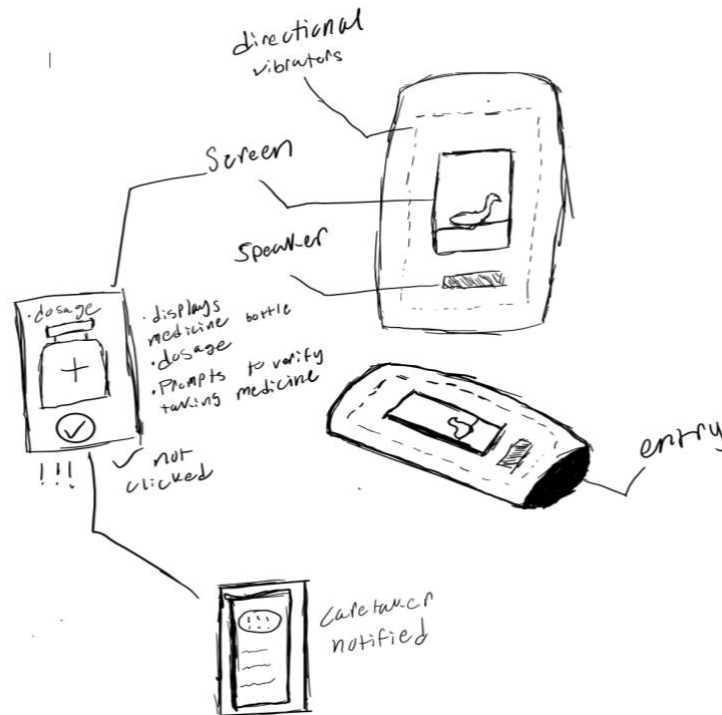
Strengths

- Requires very little interaction from the wearer after the sleeve has been put on.
- Does not require the wearer to remember to carry or regularly check a phone.
- Uses multiple forms of feedback, including visual information, audio, and vibration.
- A physical confirmation button provides a simple way to confirm medication without navigating several screens.
- Recorded family messages can make reminders more familiar and recognizable.
- Can automatically alert a caretaker when medication has not been confirmed.
- Directional vibrations provide basic guidance without requiring the wearer to interpret a map.
- May be less intimidating for users who are uncomfortable with smartphones or more complicated technology.

Weaknesses and Limitations

- The sleeve only works as intended if the wearer remembers to put it on and continues wearing it.
- The wearer may remove the sleeve, intentionally or accidentally. A sensor could potentially detect removal and alert the caretaker.
- The sleeve would require regular charging.
- The physical size and weight of the sleeve could affect the wearer's comfort.

- Pressing the confirmation button indicates that the wearer says they took the medication, but it cannot guarantee that the medication was actually taken.
- Directional vibrations may be confusing for some users and would require testing to determine whether the wearer can understand the cues.
- The system relies on the caretaker having a phone capable of receiving alerts.
- Some communication features may require Wi-Fi or another network connection.
- Entering medication information directly through a small sleeve screen may be more difficult for caretakers or doctors than using a larger interface.



Design Self-Check

What makes this design fundamentally different from the other two designs?

This design has a significantly different interaction modality from the other two designs because it is a wearable device that remains physically attached to the user. Once the sleeve is being worn, the wearer does not need to remember to carry another device or actively check an application.

The design also uses different input and output mechanisms. Rather than relying primarily on a touchscreen interface, the sleeve uses a simple display, a physical confirmation button, audio announcements, and vibrations. This could make the system more approachable for someone who is uncomfortable with technology or unwilling to navigate several screens on a phone.

What assumptions does this design make about the user and their situation?

The first major assumption is that the sleeve remains on the user. A sensor could be used to detect whether the sleeve has been removed and alert the caretaker, but the design only functions as intended while the sleeve is being worn.

Another assumption is that the caretaker has a phone capable of receiving alerts from the sleeve. The sleeve may also require access to Wi-Fi or another network connection to send these alerts. The design additionally assumes that the wearer can understand the basic information displayed on the screen, recognize the medication image, understand the audio reminders, and press the confirmation button after taking their medication.

How does the interaction flow differ from the other designs?

Users of the wearable arm sleeve rarely need to actively interact with the device. Most functions occur automatically, with the primary required interaction being a simple button press to confirm that medication has been taken. This keeps the design simple enough that users who are uncomfortable with technology may be more willing to use it.

Caretakers would not need to continuously monitor a separate application. Instead, their phone would receive an alert when the system detects an event requiring their attention, such as an unconfirmed medication or the wearer leaving a designated area. Caretakers or doctors would input medication and dosage information directly through the sleeve's screen. This information could be protected by a password to prevent accidental or intentional changes.

This interaction flow is significantly different from the other two designs because it requires relatively little input from either the wearer or caretaker while still actively monitoring the wearer. The phone application requires the dependent to navigate multiple screens and complete verification tasks, whereas the sleeve primarily displays the medication and requires a button press. The medication station focuses on medication distribution and dosage from a fixed location, whereas the sleeve remains with and actively supports the wearer.

How is this design informed by Milestone 1?

This design takes into account feedback received from Dr. Plaue regarding the potential use of vibrations for comfort and assisting individuals with Alzheimer's or other forms of memory impairment. This led the team to explore vibration as an alternative method of communicating information to the wearer rather than relying entirely on visual screens or audio alerts.

Additional feedback indicated that individuals struggling with memory may forget to use an assistive device. This concern remains present because the wearer must initially remember or be assisted in putting on the sleeve. However, once the sleeve is being worn, the wearer does not need to remember to carry, locate, or regularly check another device. The sleeve instead initiates interactions when assistance is needed.

Design 3: Smart Medication Station

Design Overview

Our Smart Medication Station is designed to stay in one permanent location in the user's home. The main idea behind this design is to make taking medication as simple as possible for someone who struggles with their memory. Instead of having the user remember which medication they need to take, the station stores the medications and gives them the correct medication at the scheduled time.

When it is time to take medication, the station lights up and plays an audio reminder. It then unlocks or presents the medication and dosage that the user is supposed to take. The station also uses sensors to tell if the medication was removed. If the user does not take the medication from the station after several reminders, a caregiver can be notified.

Medication Storage and Dispensing

The station has multiple slots that can hold different medications. Our current design has ten medication slots so that several medications can be stored in the same station. The medications would be organized based on the user's medication schedule. When it is time for a certain medication, the station only gives the user access to the medication and dosage they need at that time. The other medications would stay unavailable.

This is meant to take some of the responsibility off of the user. Instead of having to remember which bottle they need or what dosage they are supposed to take, the station handles that part of the process for them.

Lights and Speakers

The station has lights and speakers that are used to get the user's attention when it is time for medication. The lights would turn on and the speakers would play an audio reminder when a medication is ready. The speakers could also play prerecorded messages from a caregiver or family member. This gives the user the option of hearing a familiar voice instead of only hearing a regular alarm or automated message. If the medication is still not removed, the station can continue reminding the user before eventually notifying their caregiver.

Medication Detection

The station uses a sensor to detect whether the medication has been removed. This could be done using a weight sensor, lid sensor, camera, or another similar method. In our current model, we included a scale/sensor in the area where the medication is dispensed. Once the medication is removed, the station can record that the user interacted with it. This means the user does not have to press another button or go through a screen to confirm that they got their medication. One limitation is that the station can only tell

that the medication was removed. It cannot know for certain that the user actually took it. If the medication is not removed after several reminders, the station can send an alert to the caregiver.

Example Interaction

A caregiver would first load the user's medications into the station and set up their medication schedule. After that, the station would stay in the same location in the user's home. When it is time for medication, the station lights up and plays an audio reminder. A recorded message from a family member or caregiver could also play. The station then gives the user access to only the medication and dosage they need at that time.

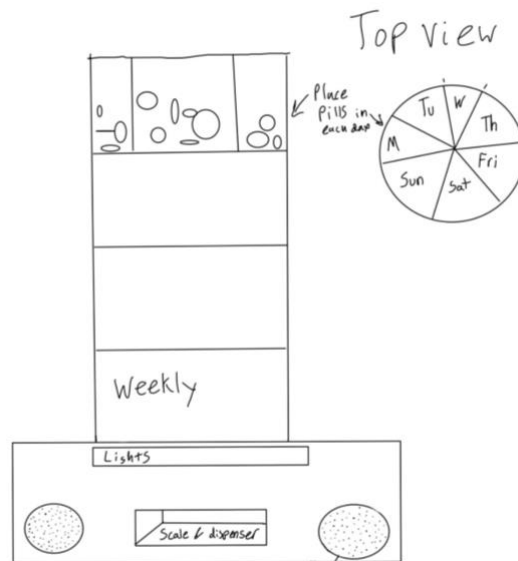
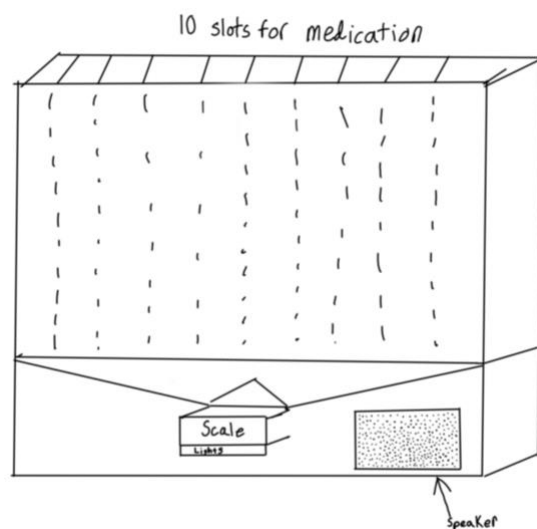
The user walks over to the station and takes the medication from the dispensing area. The sensor detects that the medication was removed and records the interaction. If the user does not get the medication, the station continues reminding them. After several reminders without the medication being removed, the caregiver receives an alert so they can check on the user.

Strengths

- The user does not have to remember which medication they are supposed to take.
- Only the correct medication and dosage are available at the scheduled time.
- Requires very little interaction from the user.
- Does not require the user to carry or know how to use a smartphone.
- Uses both lights and audio to get the user's attention.
- Can play recorded messages from family members or caregivers.
- Can detect whether medication was removed without making the user manually confirm it.
- Can alert a caregiver if medication is not removed after several reminders.
- Keeps multiple medications organized in one location.

Weaknesses and Limitations

- The station is not portable, so the user would need to be home when it is time for their medication.
- A caregiver would need to correctly load the medications and set up the schedule.
- The station can only hold a certain number or size of medications.
- Detecting that medication was removed does not guarantee that the user actually took it.
- A problem with the sensors or dispensing system could cause the station to work incorrectly.
- The user still needs to notice the reminder and be able to walk over to the station.
- The station would need to stay powered.
- Sending alerts to caregivers may require Wi-Fi or another type of network connection.
- It would probably be more expensive and complicated to create than the app or wearable sleeve.



Design Self-Check

What makes this design fundamentally different from the other two designs?

The Smart Medication Station is different from the other two designs mainly because it stays in one location and physically handles the user's medication. The phone app and wearable sleeve can move around with the user, while this design is meant to stay somewhere consistent in their home.

It also takes more responsibility away from the user. Instead of only reminding them which medication they need, the station actually controls which medication is available at that time. This means the user has fewer choices to make during the process and does not have to search for the correct medication themselves.

What assumptions does this design make about the user and their situation?

One of the biggest assumptions is that the user will be home when they need to take their medication. Since the station stays in one place, it would not be very helpful if the user was away from home when their medication was scheduled.

We also assume that there is a caregiver who can load the medication and set up the station. The user would still need to be able to notice the lights or audio reminders, walk over to the station, and take the medication that is provided. The station would also need access to power and may need an internet connection if it is going to send alerts to a caregiver.

How does the interaction flow differ from the other designs?

The Smart Medication Station requires very little direct interaction from the user. Most of the process is started by the station itself. It alerts the user, provides the correct medication, and detects when the medication is removed. The main thing the user has to do is go to the station and take the medication.

This is different from the phone app because the app requires the user to go through several screens, scan their medication, and confirm that they took it. The wearable sleeve also stays with the user throughout the day and provides reminders wherever they are. The medication station does not follow or monitor the user throughout the day and instead focuses specifically on managing medication from one location.

How is this design informed by Milestone 1?

One of the concerns that came up during Milestone 1 was that someone struggling with their memory could forget to use the system that is supposed to help them. The station tries to address this by starting the interaction itself instead of expecting the user to remember to check something. When medication is ready, the station uses lights and audio to get the user's attention.

The design also reduces how much the user has to remember about their medication. They do not have to figure out which medication or dosage they need because the station provides the correct one at the scheduled time. At the same time, the user can still take their medication independently, with the caregiver only being alerted when the system notices that they may need help.